

PETER ALEXANDER

DIRECTOR OF MATHEMATICS CURRICULUM & PRODUCT DEVELOPMENT

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Experienced mathematics curriculum and instructional product development leader specializing in scalable learning systems, digital assessment tools, and evidence-based instructional resources supporting diverse learners across K–12 and post-secondary environments. Expertise spans curriculum architecture, assessment systems, instructional technology, analytics-driven continuous improvement, and cross-functional collaboration. Skilled at building classroom-ready resources that are rigorous, engaging, teacher-usable, and aligned to standards and measurable student outcomes.

 CORE COMPETENCIES

-  Curriculum Development

 Instructional Product Design

 Assessment Design
-  Instructional Analytics

 Team Leadership

 Teacher Supports
-  Learning Progressions

 Continuous Improvement

 AI-Assisted Workflows
-  Educational Technology

 Quality Assurance

 Standards Alignment

 PROFESSIONAL EXPERIENCE

BASIS Independent Silicon Valley
Subject Expert Teacher

Fremont, CA
2024–Present

- Design and implement original mathematics curriculum for Precalculus, Honors Linear Algebra, AP Statistics, AP Physics, and interdisciplinary business analytics courses integrating conceptual understanding, modeling, and real-world applications.
- Develop digital assessment systems using HTML, JavaScript, Google Apps Script, and spreadsheet analytics to support formative assessment, mastery tracking, and rapid instructional feedback.
- Create teacher-facing instructional resources, corrective review systems, and remediation pathways aligned with evidence-based practices and student performance data.
- Leverage AI-assisted tools and analytics workflows to streamline content development, assessment generation, and instructional iteration.
- Lead DECA-aligned business and computational finance curriculum focused on data literacy, quantitative reasoning, and analytical thinking.
- Collaborate with educators and administrators to ensure curriculum coherence, standards alignment, and continuous improvement.

Illumin Education
Subject Expert Tutor & Instructional Consultant

Santa Clara, CA
2021–Present

- Provide advanced mathematics tutoring and curriculum support for middle school through college students.
- Develop individualized learning plans, assessments, and analytics dashboards to monitor progress and inform instruction.
- Build data pipelines and analytics tools using Python, R, and Google Apps Script for educational insights and reporting.
- Support families and educators with instructional strategies, college readiness, and academic planning.

Thoughtmuseum Solutions
Founder & Educational Technology Consultant

Foster City, CA
2015–Present

- Design and build web-based instructional tools and assessment platforms using HTML, JavaScript, GitHub Pages, and Google Apps Script.
- Develop supplemental mathematics curriculum resources, interactive problem sets, and analytics-enabled dashboards.
- Consult with schools and organizations on curriculum development, assessment systems, and technology integration.
- Create scalable instructional products combining rigorous mathematics content with engaging, technology-enhanced experiences.

Alameda & Laney Colleges
Mathematics Instructor

East Bay, California

- Taught college-level mathematics courses including Precalculus, Calculus, Linear Algebra, Differential Equations, and Statistics.
- Developed assessments and instructional materials supporting conceptual understanding, accessibility, and student success.
- Implemented data-informed teaching strategies to support diverse learners and improve retention.

- Taught programming, data science, and mathematics courses; supported students in advanced quantitative coursework.
- Developed curriculum and learning resources integrating Python, data analysis, and mathematical modeling.
- Contributed to educational technology projects and open-source instructional tools.

 EDUCATION

University of California, Berkeley
C.Phil., Mathematics

University of California, Berkeley
M.A., Mathematics

University of California, Berkeley
B.A., Mathematics

U.C. Berkeley Extension, Berkeley
Full-Stack Developer Program —
MERN

 TECHNICAL SKILLS & TOOLS

Languages: Python, R, JavaScript, HTML/CSS, SQL

Tools: Google Apps Script, GitHub, Pandas, NumPy, LaTeX, Excel/Google Sheets, Jupyter, Pandoc

Platforms: AWS, GitHub Pages, Google Workspace

Concepts: Data Analysis, Statistical Modeling, API Integration, Automation, Version Control, Web Development

Education: Curriculum Mapping, Learning Progressions, Assessment Design, UX Principles, Documentation

 SELECTED INSTRUCTIONAL & PRODUCT INITIATIVES

- Built AI-assisted admissions analytics pipeline processing large-scale educational data with automated extraction, feature engineering, and statistical modeling.
- Created interactive assessment platforms with auto-grading, real-time feedback, and analytics dashboards.
- Designed supplemental mathematics curricula blending conceptual understanding, real-world application, and computational thinking.
- Developed mastery-based remediation frameworks and learning progressions.